THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If A:B = 3:4 and B:C = 8:9, then A:C is: 1

(a) 27:32 **(b)** 3:9 **(c)** 2:3 **(d)** 3:5

2. 12 workers working 8 hours a day finish a job in 15 days. 10 workers working 6 hours a day will finish it in (days): 1

(a) 180 **(b)** 26 **(c)** 18 **(d)** 24

3. The ratio of boys to girls in a school is 7:8. If there are 1350 students, the number of girls is: 1

(a) 720 **(b)** 675 **(c)** 630 **(d)** 1335

4. A sum of ₹ 1200 buys 24 kg of rice at ₹ 50 per kg. When the price falls to ₹ 40 per kg, the same money buys (kg): 1

(a) 30 **(b)** 28 **(c)** 24 **(d)** 48

5. A train at 60 km/h takes 9 hours for a journey. At 45 km/h it will take (hours): 1

(a) 6.75 **(b)** 12 **(c)** 11 **(d)** 13

6. Two numbers are in the ratio 7:11 and their sum is 144. Their difference is: 1

(a) 32 **(b)** 34 **(c)** 18 **(d)** 72

7. In a pie chart of 144 students a sector has a central angle of 80°. The number of students it represents is: 1

(a) 32 **(b)** 8 **(c)** 64 **(d)** 35

8. If A:B = 2:7 and B:C = 14:5, then A:C is: 1

(a) 4:5 **(b)** 10:98 **(c)** 2:5 **(d)** 2:3

9. 9 workers working 6 hours a day finish a job in 20 days. 15 workers working 4 hours a day will finish it in (days): 1

(a) 12 **(b)** 20 **(c)** 180 **(d)** 18

10. The ratio of boys to girls in a school is 5:8. If there are 780 students, the number of girls is: 1

(a) 480 **(b)** 390 **(c)** 300 **(d)** 767

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Two numbers are in the ratio 5:8 and their sum is 143. Find the numbers. 2

12. A hostel has food for 50 students for 16 days. How long will the food last if 10 students leave? 2

13. A pie chart of 108 students has a sector of 120°. How many students does it represent, and what percent of the students is that? 2

14. Two numbers are in the ratio 7:9 and their sum is 160. Find the numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 10 workers working 6 hours a day can finish a job in 24 days. In how many days can 12 workers working 8 hours a day finish it? 3

16. A:B = 1:2 and B:C = 3:4. If A+B+C = 170, find the value of each. 3

17. In a pie chart, the sector for Maths measures 60° and represents 15 students. Find the total number of students and the number of students represented by a 96° sector. 3

OR

15 workers working 8 hours a day can finish a job in 12 days. In how many days can 18 workers working 5 hours a day finish it? 3

18. A:B = 2:3 and B:C = 3:5. If A+B+C = 200, find the value of each. 3

OR

In a pie chart, the sector for Art measures 45° and represents 12 students. Find the total number of students and the number of students represented by a 120° sector. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A mixture of 60 litres has milk and water in the ratio 2:3. How much water must be added to make the ratio 2:4? 5

OR

12 workers can complete a task in 20 days. After 8 days, 6 more workers join them. In how many more days will the work be finished? 5

20. A mixture of 84 litres has milk and water in the ratio 3:4. How much water must be added to make the ratio 3:6? 5

OR

10 workers can complete a task in 30 days. After 12 days, 8 more workers join them. In how many more days will the work be finished? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND